

Project Closeout Report

eUTxO Fundamentals: Plutarch and more [eBook]

Project number:

1400053

Project Manager:

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Date project started:

11/01/2025

Date project completed:

07/15/2026

Challenge:

F14: Cardano Open – Ecosystem (Education)

List of Challenge KPIs

1. Open Source Repository Growth:

- The existing open-source repository (55+ stars on GitHub, one of the most used resources for new Cardano developers) was extended with new chapters and fully updated material, released under the AFL-3.0 license.
- <https://github.com/elRaulito/eUTxO-Fundamentals-Building-Cardano-Smart-Contracts>

2. Documentation Quality:

- The complete eBook remains available as `light.pdf` and `dark.pdf` on the rendered branch, catering to different reading preferences and devices.
- <https://github.com/elRaulito/eUTxO-Fundamentals-Building-Cardano-Smart-Contracts/blob/rendered/dark.pdf>

3. New Educational Content:

- **Plutarch** (4.6 Plutarch: Advanced Functional Contract Composition), covering advanced functional contract composition on Cardano.
- **Marketplace contracts** extended with Plutarch implementations, testing and deployment guides (7.2, 7.3).
- **Governance on Cardano** (Chapter 8): on-chain governance, governance transactions, governance smart contracts and real-world use cases.
- **Advanced Techniques for Cardano Smart Contracts** (Chapter 9): linked lists as infinite arrays in eUTxO, merklized validators for scalable contracts, and special tips and tricks.
- **Updated material:** obsolete off-chain code based on the deprecated Lucid library was migrated to the actively maintained MeshJS (@meshsdk/core) across all text, examples and exercise solutions.

4. Community Engagement:

- Releases were announced and discussed on social media, and community feedback via GitHub issues and pull requests was incorporated throughout the project.
- <https://x.com/elraulito/status/2077319468703752467>

List of Project KPIs

1. Clear Project Scope and Objectives:

- Extend the most-used onboarding eBook for Cardano developers with Plutarch and advanced chapters, and bring the existing material up to date – from setting up a node and connecting a wallet, up to building a complete NFT marketplace in multiple smart contract languages.

2. Milestone Delivery:

- All milestones were completed: Governance on Cardano, Plutarch, Advanced tricks for smart contracts, and the Final eBook covering all ten chapters.

3. Working Code Examples:

- All new chapters ship with working, open-source code examples (Plutarch validators, governance transactions with MeshJS, marketplace contracts) that developers can copy, run and build upon.

4. Code Optimization and Maintenance:

- The codebase and text were continuously improved through community pull requests, and deprecated tooling was replaced to keep the book aligned with the current Cardano developer stack.

Key Achievements

- **Advanced Content Made Accessible:** The Plutarch, governance and advanced-techniques chapters bring production-grade smart contract patterns into a beginner-accessible resource, with working code examples for each topic.
- **Complete, Up-to-date eBook:** The book now covers the full journey – environment setup, the eUTxO model, six smart contract languages including Plutarch, runtime and execution, a full marketplace project, governance, and advanced techniques.
- **Community Feedback:** Feedback through GitHub issues, pull requests, Discord and Twitter has been instrumental in refining the new chapters and identifying outdated material.

Key Learnings

- **Keeping Material Current:** Documentation in a fast-moving ecosystem decays quickly; part of the project effort went into replacing deprecated libraries and outdated instructions, which proved as valuable to readers as the new content itself.
- **Advanced Content Needs Extra Care:** Covering advanced patterns (merklized validators, linked lists in eUTxO) correctly requires deep research and validation against real-world production contracts, which took a significant share of the project effort.
- **Community Engagement:** Regular interaction with the community remains the key driver for improvements, surfacing issues and generating ideas for new content.
- **Continuous Improvement:** The open-source nature of the project allows the content to evolve through community contributions and stay aligned with the latest developments in the Cardano ecosystem.

Next Steps for the Product or Service Developed

- **Maintenance:** Keep the book aligned with new protocol features and tooling releases, with community pull requests reviewed regularly.

- **Expansion of Features:** Additional use cases and advanced topics will be added, driven by community requests and contributions.
- **Promotion and Outreach:** Continued sharing of chapters and updates on social media to onboard new developers from other ecosystems to Cardano.

Final Thoughts/Comments

With the Plutarch, governance and advanced-techniques chapters completed and all legacy material updated, the eUTxO Fundamentals eBook is now a complete, current and fully open-source path from first wallet connection to advanced smart contract patterns on Cardano. The open-source nature of the project and the support from the community will help it evolve and grow as a valuable resource for developers interested in the Cardano ecosystem.

Links to other relevant project sources or documents

- GitHub Repository: [https://github.com/elRaulito/eUTxO-Fundamentals-Building-Cardano-Smart-Contracts](https://github.com/elRaulito/eUTxO-Fundamentals-Building-Cardano-Smart-Contracts/blob/main/eUTxO-Fundamentals-Building-Cardano-Smart-Contracts.pdf)
- Rendered PDF: [https://github.com/elRaulito/eUTxO-Fundamentals-Building-Cardano-Smart-Contracts/blob/main/eUTxO-Fundamentals-Building-Cardano-Smart-Contracts/blob/rendered/dark.pdf](https://github.com/elRaulito/eUTxO-Fundamentals-Building-Cardano-Smart-Contracts/blob/main/eUTxO-Fundamentals-Building-Cardano-Smart-Contracts.pdf)
- Twitter Announcement: <https://x.com/elraulito/status/2077319468703752467>
- Fund Project URL: <https://projectcatalyst.io/funds/14>, Project ID #1400053
- Closeout Video: <https://youtu.be/gFWVOYIDMz8>